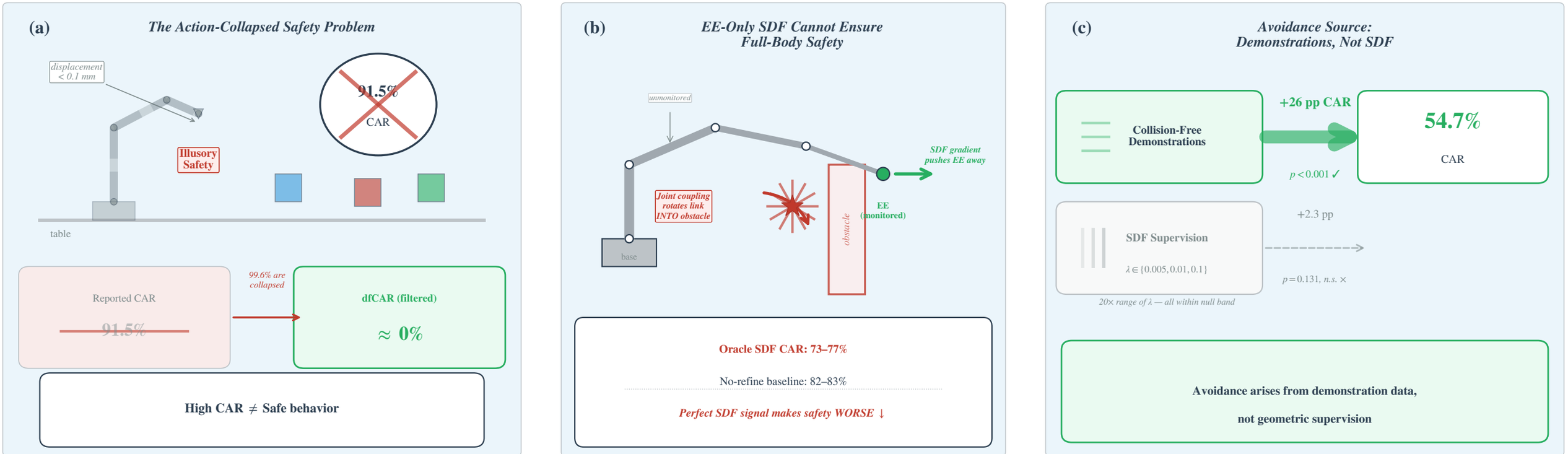
(a)
The Action-Collapsed Safety Problem
displacement
< 0.1 mm
Illusory
Safety
91.5%
CAR
table
Reported CAR
91.5%
99.6% are
collapsed
dfCAR (filtered)
≈ 0%
High CAR ≠ Safe behavior
(b)
EE-Only SDF Cannot Ensure
Full-Body Safety
unmonitored
Joint coupling
rotates link
INTO obstacle
SDF gradient
pushes EE away
EE
(monitored)
obstacle
base
Oracle SDF CAR: 73–77%
No-refine baseline: 82–83%
Perfect SDF signal makes safety WORSE ↓
(c)
Avoidance Source:
Demonstrations, Not SDF
Collision-Free
Demonstrations
+26 pp CAR
54.7%
CAR
p < 0.001 ✓
SDF Supervision
λ ∈ {0.005, 0.01, 0.1}
+2.3 pp
p = 0.131, n.s. ×
20× range of λ — all within null band
Avoidance arises from demonstration data,
not geometric supervision